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INTRODUCTION

1.1 Background

The rapid growth of information technology has transformed the way individuals, businesses, educational institutions, and government organizations manage, process, and exchange information. Modern organizations rely heavily on interconnected computer networks, cloud services, and internet-based applications to perform daily operations efficiently. While these technological advancements have significantly improved productivity and accessibility, they have also increased the number and complexity of cyber threats. Attackers continuously attempt to exploit vulnerabilities in operating systems, networks, and applications to gain unauthorized access, steal sensitive information, disrupt services, or compromise system integrity.

To address these growing cybersecurity challenges, organizations increasingly adopt **Security Information and Event Management (SIEM)** solutions for centralized security monitoring and threat detection. SIEM platforms collect and analyze logs from multiple devices and systems, enabling security teams to detect suspicious activities, investigate security incidents, and respond to threats in real time.

This project focuses on the design and implementation of a Home Network Security Information and Event Management (SIEM) system using Wazuh and pfSense in a virtualized environment. The implementation integrates Wazuh SIEM, pfSense Firewall, Windows 10 Endpoint, and Kali Linux using Oracle VirtualBox to create a practical security monitoring laboratory. Wazuh collects and analyzes logs generated from the Windows endpoint and network devices, while pfSense provides network security and traffic management. Attack simulations performed using Kali Linux generate security events that are collected, analyzed, and visualized through the Wazuh Dashboard.

By integrating these technologies, the project demonstrates how organizations can perform centralized log management, continuous security monitoring, threat detection, incident analysis, and threat hunting using open-source cybersecurity tools. The project also provides practical experience in Security Operations Center (SOC) activities and helps students develop skills in log analysis, event correlation, digital forensics, and cyber threat detection.

1.2 Problem Statement

As organizations become increasingly dependent on digital infrastructure, cyberattacks have become more frequent, sophisticated, and difficult to detect. Attackers continuously perform network reconnaissance, brute-force login attempts, privilege escalation, malware execution, unauthorized access, and other malicious activities targeting vulnerable systems. Traditional security mechanisms such as antivirus software and firewalls provide an important layer of defense but often lack centralized visibility into security events occurring across multiple systems.

One of the major challenges faced by cybersecurity professionals is the timely detection and analysis of security incidents from large volumes of log data generated by different operating systems, applications, and network devices. Without centralized monitoring and correlation of security events, organizations may fail to identify ongoing attacks or respond effectively to security incidents.

This project addresses these challenges by implementing a centralized SIEM solution using Wazuh integrated with pfSense in a virtualized home network environment. The proposed solution collects

logs from Windows endpoints, analyzes security events, detects suspicious activities such as failed login attempts and file modifications, maps alerts to the MITRE ATT&CK framework, and provides real-time dashboards for security monitoring and incident analysis. The implementation enables organizations and students to understand modern SOC operations while improving visibility into network and endpoint security events.

1.3 Objectives

The primary objective of this project is to design and implement a Home Network Security Information and Event Management (SIEM) system using Wazuh and pfSense for real-time security monitoring, threat detection, log analysis, and incident response.

The specific objectives of the project are:

- To understand the concepts and importance of Security Information and Event Management (SIEM) in modern cybersecurity.
 - To deploy and configure the Wazuh SIEM platform in a virtual environment.
 - To configure pfSense as the network firewall for communication between virtual machines.
 - To integrate a Windows 10 endpoint with the Wazuh server for centralized log collection.
 - To monitor Windows security events, system logs, and file integrity changes.
 - To perform attack simulations using Kali Linux to generate security events.
 - To detect failed login attempts, user account creation and deletion, and network reconnaissance.
 - To analyze alerts using the Wazuh Dashboard and correlate them with the MITRE ATT&CK framework.
 - To perform threat hunting using centralized security event analysis.
 - To provide a practical Security Operations Center (SOC) laboratory for cybersecurity learning and research.
-

1.4 Scope of the Project

The scope of this project is focused on designing and implementing a secure and centralized SIEM environment for monitoring security events, detecting cyber threats, and analyzing system activities within a virtual home network. The project integrates Wazuh, pfSense, Windows 10, and Kali Linux using Oracle VirtualBox to simulate a realistic enterprise security monitoring environment.

The implemented system demonstrates various SOC functions, including centralized log collection, endpoint monitoring, File Integrity Monitoring (FIM), attack simulation, threat hunting, and MITRE ATT&CK mapping. Wazuh vulnerability information was available as a platform capability but was not treated as a primary experimental detection result. The project is intended primarily for educational purposes, cybersecurity research, and practical SOC training.

However, the project has certain limitations. The implementation is deployed within a virtual laboratory and does not represent a large-scale enterprise infrastructure. While Wazuh provides comprehensive monitoring and detection capabilities, the system does not include advanced Security Orchestration, Automation, and Response (SOAR) features, cloud-native monitoring, or commercial threat intelligence integrations. Therefore, the project should be considered a learning and research platform rather than a complete enterprise security solution.

LITERATURE SURVEY

2.1 Introduction

Cybersecurity has become one of the most critical aspects of modern information technology due to the increasing frequency and sophistication of cyber threats. Organizations across various sectors face continuous risks from cyberattacks such as malware infections, ransomware, phishing, brute-force attacks, insider threats, privilege escalation, and unauthorized access attempts. These attacks can lead to financial losses, data breaches, service disruptions, and reputational damage.

To address these challenges, organizations deploy multiple security solutions such as firewalls, antivirus software, Intrusion Detection Systems (IDS), Intrusion Prevention Systems (IPS), Endpoint Detection and Response (EDR), and Security Information and Event Management (SIEM) platforms. Among these technologies, SIEM has become an essential component of modern cybersecurity because it enables centralized log collection, real-time monitoring, event correlation, and incident response.

A literature survey helps understand the evolution of SIEM technologies, their advantages and limitations, and the role of open-source platforms such as Wazuh in modern Security Operations Centers (SOCs). It also provides insight into firewall technologies like pfSense and their contribution to network security and threat detection.

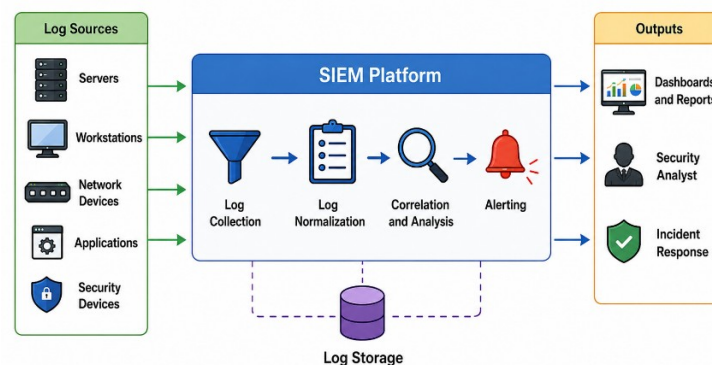


Figure 2.1 Security Information and Event Management (SIEM) Architecture

2.2 Security Information and Event Management (SIEM)

Security Information and Event Management (SIEM) is a cybersecurity solution that combines Security Information Management (SIM) and Security Event Management (SEM) to provide centralized security monitoring and analysis. SIEM platforms collect log data from multiple sources, including operating systems, network devices, servers, applications, and security appliances. These logs are normalized, correlated, and analyzed to identify suspicious activities and potential security incidents.

Modern SIEM solutions enable security analysts to detect threats in real time, investigate incidents, generate compliance reports, and perform threat hunting activities. They also support automated alert generation, event correlation, and integration with threat intelligence feeds, helping organizations improve their overall cybersecurity posture.

The adoption of SIEM solutions has become increasingly important as organizations generate large volumes of security logs that cannot be analyzed manually. By providing centralized visibility into security events, SIEM platforms significantly reduce incident response time and improve the efficiency of Security Operations Centers (SOCs).

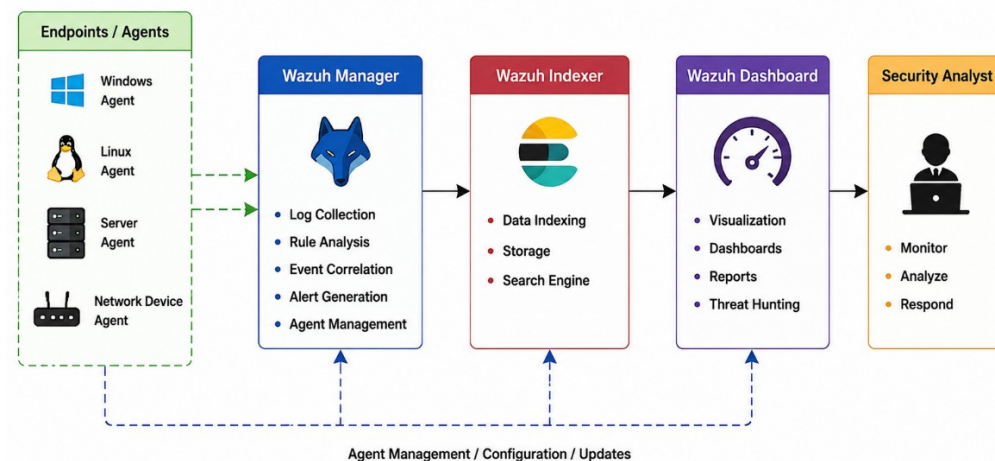


Figure 2.2 Wazuh SIEM Components

2.3 Wazuh Platform

Wazuh is an open-source Security Information and Event Management (SIEM) and Extended Detection and Response (XDR) platform designed to provide comprehensive security monitoring and threat detection. It collects security logs from endpoints, servers, network devices, and cloud environments through lightweight agents installed on monitored systems.

The Wazuh platform consists of several key components, including the Wazuh Manager, Wazuh Indexer, Wazuh Dashboard, and Wazuh Agents. The Wazuh Manager analyzes security events and applies detection rules, while the Indexer stores collected data for efficient searching and analysis. The Dashboard provides a web-based interface for monitoring alerts, performing threat hunting, and visualizing security events.

Key features of Wazuh include:

- Log collection and analysis
- File Integrity Monitoring (FIM)
- Vulnerability detection
- Security configuration assessment
- Malware detection

- Threat hunting
- MITRE ATT&CK mapping
- Compliance monitoring

These capabilities make Wazuh a powerful and cost-effective solution for organizations seeking enterprise-level security monitoring using open-source technologies.

Table 2.1: Comparison of SIEM Solutions

SIEM Solution	Type	Advantages	Limitations
Wazuh	Open Source	Free, scalable, FIM, vulnerability detection, MITRE ATT&CK mapping	Requires configuration and administration
Splunk Enterprise Security	Commercial	Advanced analytics, powerful search capabilities	High licensing cost
IBM QRadar	Commercial	Enterprise-grade threat detection and compliance support	Complex deployment and expensive licensing
Microsoft Sentinel	Cloud-based	Cloud-native SIEM with AI-powered analytics	Best suited for Microsoft Azure environments

2.4 pfSense Firewall

pfSense is an open-source firewall and router platform based on FreeBSD. It provides comprehensive network security features, including firewall filtering, Network Address Translation (NAT), Virtual Private Network (VPN) support, traffic shaping, intrusion detection integration, and network monitoring.

In this project, pfSense functions as the network gateway and firewall, controlling traffic between the virtual machines while providing secure communication among the Wazuh Server, Windows 10 endpoint, and Kali Linux attack machine. It also supports network segmentation, helping create a realistic enterprise security environment for testing and monitoring cyber threats.

The flexibility, reliability, and extensive security features of pfSense make it one of the most widely adopted open-source firewall solutions in both educational laboratories and enterprise environments.

2.5 Related Work

Several researchers and cybersecurity professionals have demonstrated the effectiveness of SIEM platforms in improving organizational security through centralized monitoring and threat detection. Commercial SIEM solutions such as Splunk Enterprise Security, IBM QRadar,

ArcSight, and Microsoft Sentinel have been widely adopted for enterprise security operations due to their advanced analytics and incident response capabilities.

In recent years, open-source SIEM platforms such as Wazuh have gained significant popularity because they provide enterprise-grade security monitoring without the high licensing costs associated with commercial products. Wazuh integrates multiple security functions, including log management, endpoint monitoring, vulnerability detection, file integrity monitoring, compliance assessment, and MITRE ATT&CK mapping within a unified platform.

This project builds upon these advancements by implementing a Home Network SIEM environment using Wazuh and pfSense in Oracle VirtualBox. The implementation demonstrates how centralized log collection, attack simulation, threat detection, and incident analysis can be performed in a virtual laboratory, providing a practical learning platform for cybersecurity education and Security Operations Center (SOC) training.

SYSTEM ANALYSIS

3.1 Existing System

Traditional cybersecurity solutions such as firewalls, antivirus software, Intrusion Detection Systems (IDS), and Intrusion Prevention Systems (IPS) are widely used to protect computer networks against cyber threats. These security mechanisms help prevent unauthorized access, detect known attack patterns, and secure network communication. However, most traditional security tools operate independently and generate large volumes of security logs, making it difficult for security administrators to obtain a centralized view of security events.

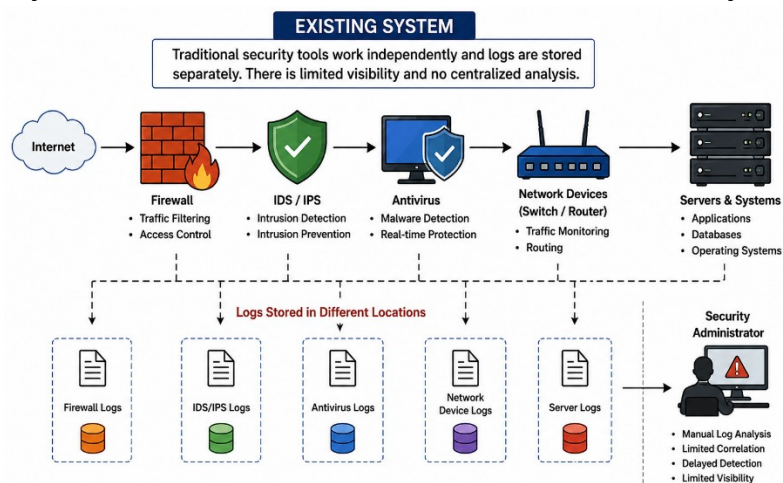


Figure 3.1 Existing System

Figure 3.1 Existing System

In many organizations, security logs generated by operating systems, applications, and network devices are stored separately without centralized correlation or analysis. This makes incident investigation more time-consuming and increases the possibility of missing critical security events. As cyber threats continue to evolve, organizations require centralized security monitoring solutions capable of collecting, correlating, and analyzing events from multiple sources in real time.

Limitations of the Existing System

- Security logs are distributed across multiple devices.
- Limited centralized visibility into security events.
- Difficult to correlate events from different systems.
- Delayed incident detection and response.
- High volume of manual log analysis.
- Limited support for threat hunting.
- Increased possibility of overlooking critical security alerts.

Table 3.1: Existing System vs Proposed System

Existing System	Proposed SIEM System
Individual security tools operate independently	Centralized security monitoring platform
Logs stored on separate devices	Centralized log collection
Manual log analysis	Automated event correlation
Limited security visibility	Real-time dashboards and alerts
Difficult incident investigation	Centralized incident analysis
Limited threat hunting capabilities	Integrated threat hunting and MITRE ATT&CK mapping

3.2 Proposed System

The proposed system implements a **Home Network Security Information and Event Management (SIEM)** solution using **Wazuh** and **pfSense** within a virtualized environment. The implementation integrates a Wazuh Server, Windows 10 endpoint, Kali Linux attack machine, and pfSense firewall using Oracle VirtualBox to create a centralized security monitoring laboratory.

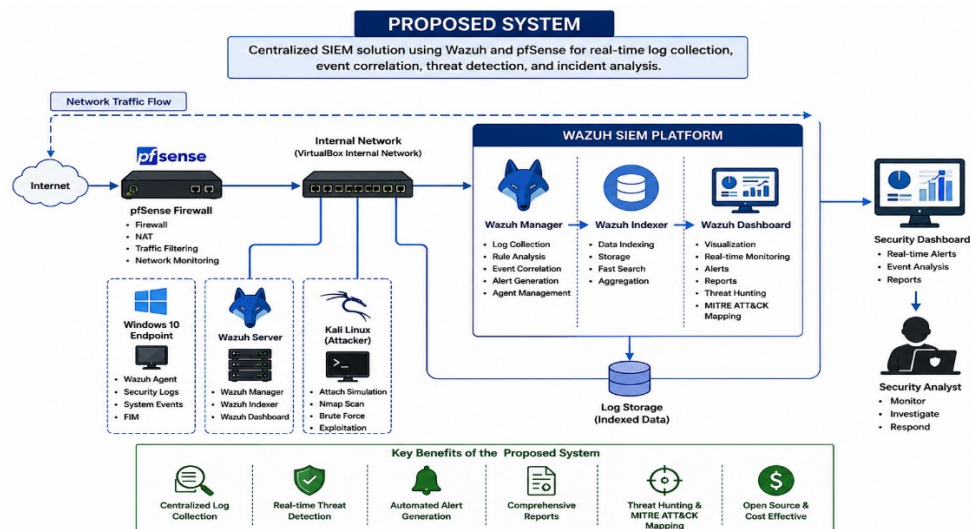


Figure 3.2 Proposed System

Figure 3.2 Proposed System

Advantages of the Proposed System

- Centralized log collection and analysis.
- Real-time security monitoring.
- Automated threat detection.
- File Integrity Monitoring (FIM).
- MITRE ATT&CK mapping.
- Threat hunting capabilities.
- Open-source and cost-effective implementation.
- Practical Security Operations Center (SOC) learning environment.

3.3 Feasibility Study

Before implementing the proposed SIEM solution, it is important to evaluate its technical, economic, operational, and schedule feasibility. The project has been analyzed based on these factors to ensure successful deployment and effective security monitoring.

Table 3.2: Feasibility Analysis

Feasibility Type	Description
Technical Feasibility	The system is implemented using Oracle VirtualBox, Wazuh, pfSense, Windows 10, and Kali Linux, which are stable, reliable, and widely used cybersecurity technologies.
Economic Feasibility	The project primarily uses open-source software, significantly reducing implementation and maintenance costs.
Operational Feasibility	The Wazuh Dashboard provides a user-friendly interface for centralized security monitoring, alert analysis, and threat hunting.
Schedule Feasibility	The project was successfully implemented and tested within the planned project schedule.

3.3.1 Technical Feasibility

The proposed system is technically feasible because it utilizes mature and widely adopted open-source technologies such as Wazuh, pfSense, Oracle VirtualBox, Windows 10, and Kali Linux. These technologies provide reliable security monitoring, endpoint protection, network traffic control, and attack simulation capabilities. Their extensive documentation and active community support simplify deployment, configuration, and maintenance.

3.3.2 Economic Feasibility

The proposed solution is economically feasible because most of the software components used in the project are open source and freely available. Organizations and educational institutions can implement similar SIEM environments without purchasing expensive commercial security solutions. Only standard computer hardware with sufficient processing power and memory is required to deploy the virtual machines, making the project affordable for learning, research, and small-scale security operations.

CHAPTER 4

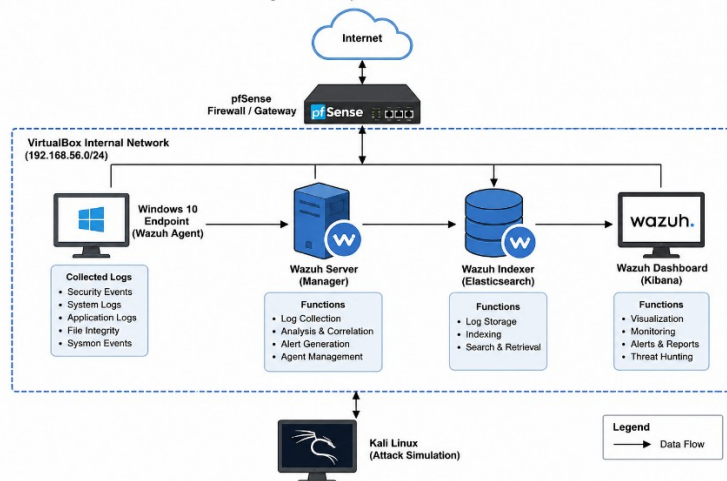
SYSTEM DESIGN

4.1 Introduction

The proposed Home Network Security Information and Event Management (SIEM) system is designed to provide centralized security monitoring, log collection, threat detection, and incident analysis within a virtualized environment. The system integrates Wazuh SIEM, pfSense Firewall, Windows 10 Endpoint, and Kali Linux using Oracle VirtualBox to simulate an enterprise security monitoring infrastructure.

The architecture enables security events generated from the Windows endpoint and network activities to be collected, analyzed, and visualized through the Wazuh Dashboard. The design provides a practical virtual platform for learning Security Operations Center (SOC) operations, threat hunting, and incident analysis.

Figure 4.1: System Architecture

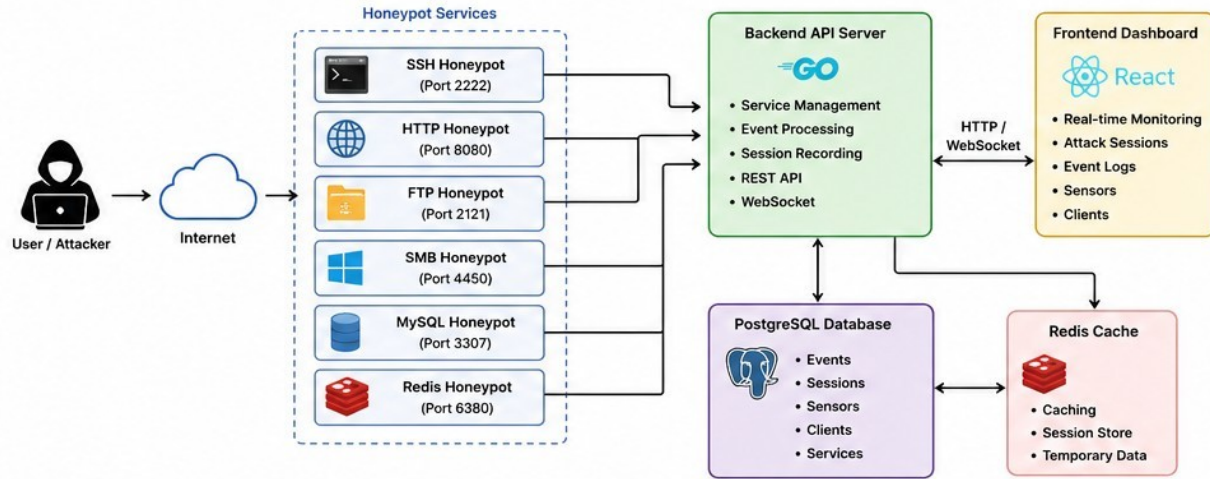


4.2 System Architecture

The proposed SIEM system follows a layered architecture consisting of the following components:

1. **Network Security Layer** – Provides firewall protection and network traffic management using pfSense.
2. **Endpoint Layer** – Consists of the Windows 10 endpoint monitored by the Wazuh Agent.
3. **Security Monitoring Layer** – Includes the Wazuh Manager, Wazuh Indexer, and Wazuh Dashboard for centralized security monitoring.
4. **Attack Simulation Layer** – Uses Kali Linux to generate security events through controlled attack simulations.

Figure 4.1: Overall System Architecture



4.3 System Components

4.3.1 Wazuh Server

The Wazuh Server is the central component of the proposed SIEM system. It consists of the Wazuh Manager, Wazuh Indexer, and Wazuh Dashboard.

Its responsibilities include:

- Collecting logs from monitored endpoints.
- Event correlation and rule analysis.
- Alert generation.
- Threat detection.
- Data indexing and storage.
- Dashboard visualization.
- Threat hunting.

4.3.2 pfSense Firewall

pfSense functions as the network firewall and gateway for the virtual environment.

Its responsibilities include:

- Network traffic filtering.
- Firewall rule enforcement.
- Network Address Translation (NAT).
- Secure communication between virtual machines.
- Network segmentation.

4.3.3 Windows 10 Endpoint

The Windows 10 virtual machine acts as the monitored endpoint within the SIEM environment.

The installed Wazuh Agent collects:

- Windows Event Logs.
- Security Events.
- System Logs.
- File Integrity Monitoring (FIM) events.
-

These logs are forwarded to the Wazuh Server for analysis.

4.3.4 Kali Linux Attack Machine

Kali Linux is used as the attack simulation platform for generating security events.

Typical activities include:

- Nmap Network Scanning.
- Failed Login Attempts.
-
- File Modification Testing.
- User Account Creation and Deletion.
- Network Reconnaissance.

These activities generate alerts that are analyzed by Wazuh.

4.4 Virtual Network Design

The complete security monitoring environment is deployed using Oracle VirtualBox.

The virtual network consists of:

- pfSense Firewall
- Wazuh Server
- Windows 10 Endpoint
- Kali Linux Attack Machine

All virtual machines communicate through an Internal Network (LAN), while pfSense provides controlled Internet access using a NAT interface.

Table 4.1: Virtual Machine Configuration

Virtual Machine	Operating System	Purpose
pfSense	FreeBSD-based	Firewall and Router
Wazuh Server	Ubuntu Server	SIEM Platform
Windows 10	Windows 10	Monitored Endpoint
Kali Linux	Kali Linux	Attack Simulation

Figure 4.2: Virtual Machine Network Configuration

Figure 4.2: Virtual Machine Network Configuration

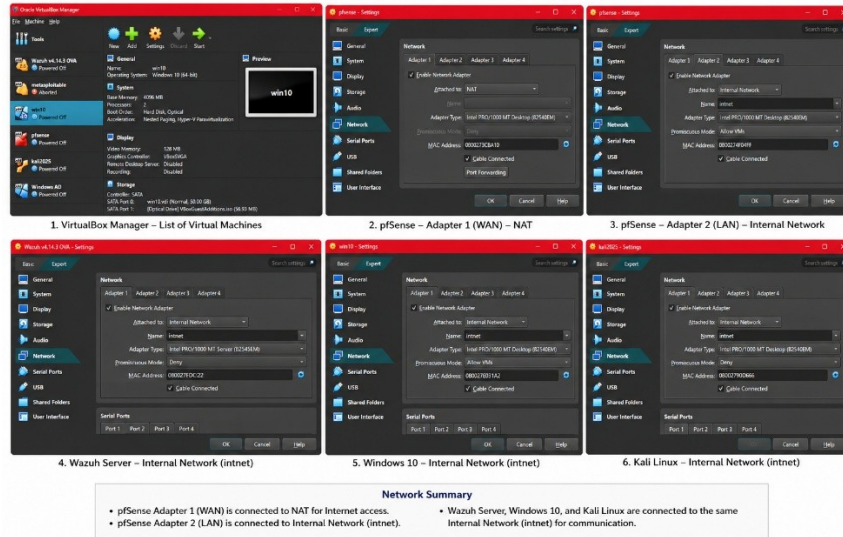


Table 4.2: Software and System Components

Software / Component	Version / Platform	Purpose
Oracle VirtualBox	7.x	Virtualization platform for hosting virtual machines
Wazuh Server	4.14.3	Centralized SIEM platform for log collection and threat detection
pfSense	CE	Firewall, router, and network traffic management
Windows 10	64-bit	Monitored endpoint generating security events
Kali Linux	2025	Attack simulation and security testing
Wazuh Agent	4.14.3	Collects endpoint logs and forwards them to the Wazuh Server
Sysmon	Latest	Advanced Windows event logging for enhanced monitoring

4.5 Security Monitoring Design

The proposed SIEM system is designed to provide centralized security monitoring by integrating Wazuh, pfSense, Windows 10, and Kali Linux within a virtualized environment. The Wazuh Agent installed on the Windows 10 endpoint continuously collects system logs, Windows Security Events, and File Integrity Monitoring (FIM) events. These logs are forwarded to the Wazuh Manager for analysis.

The Wazuh Manager applies predefined detection rules to identify suspicious activities such as failed login attempts, user account creation and deletion, network reconnaissance, and file

modifications. The processed events are stored in the Wazuh Indexer and visualized through the Wazuh Dashboard, enabling real-time monitoring, threat hunting, and incident analysis.

4.6 Data Flow Process

The working flow of the proposed SIEM system is as follows:

1. User activities or attack simulations generate security events on the Windows 10 endpoint.
2. The Wazuh Agent collects security logs and system events from the monitored endpoint.
3. The collected logs are forwarded to the Wazuh Manager for analysis.
4. The Wazuh Manager correlates events, applies detection rules, and generates security alerts.
5. The processed events are indexed and stored in the Wazuh Indexer.
6. The Wazuh Dashboard retrieves the indexed data and displays alerts, dashboards, and reports.
7. Security analysts perform threat hunting and incident analysis based on the generated alerts.

CHAPTER 5

IMPLEMENTATION

5.1 Implementation Environment

The Home Network Security Information and Event Management (SIEM) system was implemented on a Windows 11 host machine using Oracle VirtualBox as the virtualization platform. The virtual environment consists of four primary virtual machines: pfSense Firewall, Wazuh Server, Windows 10 Endpoint, and Kali Linux. This setup provides an isolated laboratory for centralized security monitoring, attack simulation, and threat analysis.

The Wazuh Server was deployed on Ubuntu Server and configured with the Wazuh Manager, Wazuh Indexer, and Wazuh Dashboard. The Windows 10 virtual machine was configured as the monitored endpoint with the Wazuh Agent and Sysmon installed to support enhanced Windows event logging. pfSense was configured as the firewall and router, providing network traffic filtering and communication between the virtual machines. Kali Linux was used to perform controlled attack simulations, including network scanning, failed login attempts, and other security testing activities.

All virtual machines communicate through an Internal Network configured in Oracle VirtualBox, while pfSense provides Internet connectivity using a Network Address Translation (NAT) interface. Security events generated by the Windows endpoint are collected by the Wazuh Agent, analyzed by the Wazuh Manager, stored in the Wazuh Indexer, and visualized through the Wazuh Dashboard. This implementation provides a practical Security Operations Center (SOC) environment for log analysis, threat detection, incident response, and threat hunting using open-source cybersecurity technologies.

Table 5.1: Software Environment

Software / Tool	Version	Purpose
Windows 11	24H2 / 25H2	Host Operating System
Oracle VirtualBox	7.1.4	Virtualization Platform
Wazuh Server	4.14.3	Security Information and Event Management (SIEM)
Ubuntu Server	24.04 LTS	Operating System for Wazuh Server
pfSense	pfSense CE 2.8.0	Firewall and Router
Windows 10	22H2	Monitored Endpoint
Kali Linux	Kali Linux 2025	Attack Simulation and Security Testing
Wazuh Agent	4.14.3	Endpoint Log Collection
Sysmon	Microsoft Sysmon	Advanced Windows Event Logging

5.2 Deployment Process

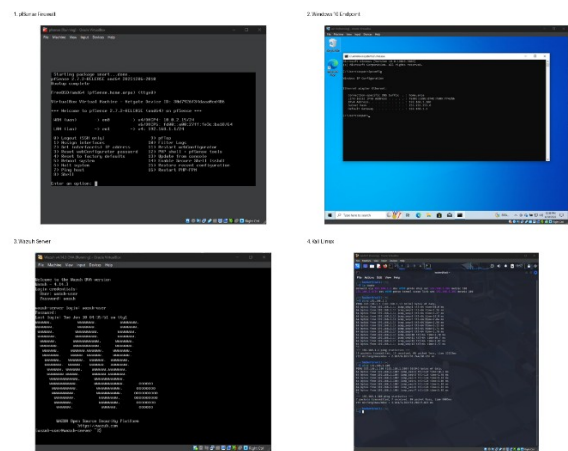
The implementation of the Home Network Security Information and Event Management (SIEM) system began by creating a virtualized environment using Oracle VirtualBox. Four virtual machines were deployed: **pfSense Firewall**, **Wazuh Server**, **Windows 10 Endpoint**, and **Kali Linux**. These machines were connected through an Internal Network, while pfSense provided Internet access using Network Address Translation (NAT).

The Wazuh Server was installed on Ubuntu Server and configured with the Wazuh Manager, Wazuh Indexer, and Wazuh Dashboard. The Windows 10 endpoint was configured with the Wazuh Agent and Sysmon to collect and forward security events to the Wazuh Server.

During deployment, issues such as Wazuh Indexer failures, agent connectivity problems, and network communication errors were encountered. These were resolved by verifying service status, correcting network settings, and restarting the required services. After successful agent registration, the Windows endpoint began forwarding logs to the Wazuh Server.

Finally, Kali Linux was used to perform attack simulations such as network scanning, failed login attempts, and file modifications. The documented events were detected, analyzed, and displayed on the Wazuh Dashboard for security monitoring and threat analysis.

Figure 5.1: Oracle VirtualBox Environment



5.3 Wazuh Service Initialization

After deployment, the Wazuh Manager, Wazuh Indexer, and Wazuh Dashboard services were started and verified to be running successfully. The Wazuh Agent was enrolled on the Windows 10 endpoint, enabling the collection and forwarding of security events for real-time monitoring.

Table 5.2: Implemented Wazuh Services

Service	Port	Function
Wazuh Manager	1514, 1515	Receives logs and manages Wazuh agents

Service	Port	Function
Wazuh Dashboard	443	Provides web interface for security monitoring
Wazuh Indexer	9200	Stores and indexes security events
Wazuh Agent	1514, 1515	Collects and forwards endpoint logs
pfSense Firewall	LAN/WAN	Provides routing, firewall, and network security
Sysmon	Windows Service	Generates detailed Windows security events

Figure 5.2: Wazuh Services Initialization

```

SEC/APP/TUN...
(wazuh-user@wazuh-server ~)$ sudo systemctl restart wazuh-manager
(wazuh-user@wazuh-server ~)$ sudo systemctl status wazuh-manager
● wazuh-manager.service - Wazuh manager
   Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-06-30 09:52:40 UTC; 4min 5s ago
     Process: 2486 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
    Main PID: 2486 (code=exited, status=0/SUCCESS)
      Tasks: 244 (limit: 5173)
     Memory: 867.6M
        CPU: 3min 17.486s
   CGroup: /system.slice/wazuh-manager.service
           └─2347 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             2348 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             2349 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             2352 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             2355 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             2585 /var/ossec/bin/wazuh-authd
             2605 /var/ossec/bin/wazuh-db
             2645 /var/ossec/bin/wazuh-execd
             2664 /var/ossec/bin/wazuh-analysisd
             2682 /var/ossec/bin/wazuh-syscheckd
             2704 /var/ossec/bin/wazuh-remoted
             2732 /var/ossec/bin/wazuh-logcollector
             2806 /var/ossec/bin/wazuh-monitord
             2827 /var/ossec/bin/wazuh-modulesd

Jun 30 09:52:37 wazuh-server env(2486): wazuh-logcollector: Process 2750 not used by Wazuh, removing...
Jun 30 09:52:37 wazuh-server env(2486): Started wazuh-logcollector...
Jun 30 09:52:37 wazuh-server env(2486): wazuh-monitord: Process 2818 not used by Wazuh, removing...
Jun 30 09:52:37 wazuh-server env(2486): Started wazuh-monitord...
Jun 30 09:52:37 wazuh-server env(2486): wazuh-modulesd: Process 2839 not used by Wazuh, removing...
Jun 30 09:52:37 wazuh-server env(2825): 2026-06-30 09:52:37 wazuh-modulesd:router: INFO: Loaded router
Jun 30 09:52:37 wazuh-server env(2825): 2026-06-30 09:52:37 wazuh-modulesd:content_manager: INFO: Loaded content manager
Jun 30 09:52:37 wazuh-server env(2825): 2026-06-30 09:52:37 wazuh-modulesd:inventory-harvester: INFO: Loaded inventory harvester
Jun 30 09:52:38 wazuh-server env(2486): Started wazuh-modulesd...
Jun 30 09:52:40 wazuh-server env(2486): Completed.
Times 1-33/33 (END)

```

5.4 Dashboard Monitoring and Testing

After all Wazuh services were successfully initialized, the **Wazuh Dashboard** was accessed through a web browser. The dashboard displayed the registered endpoint, security events, alerts, and system status in real time.

To verify the implementation, the Windows 10 endpoint was connected to the Wazuh Server using the Wazuh Agent. Attack simulations performed from the Kali Linux virtual machine generated security events, which were successfully detected and displayed on the Wazuh Dashboard, confirming the proper functioning of the SIEM environment.

Figure 5.3: Wazuh Dashboard Home Page

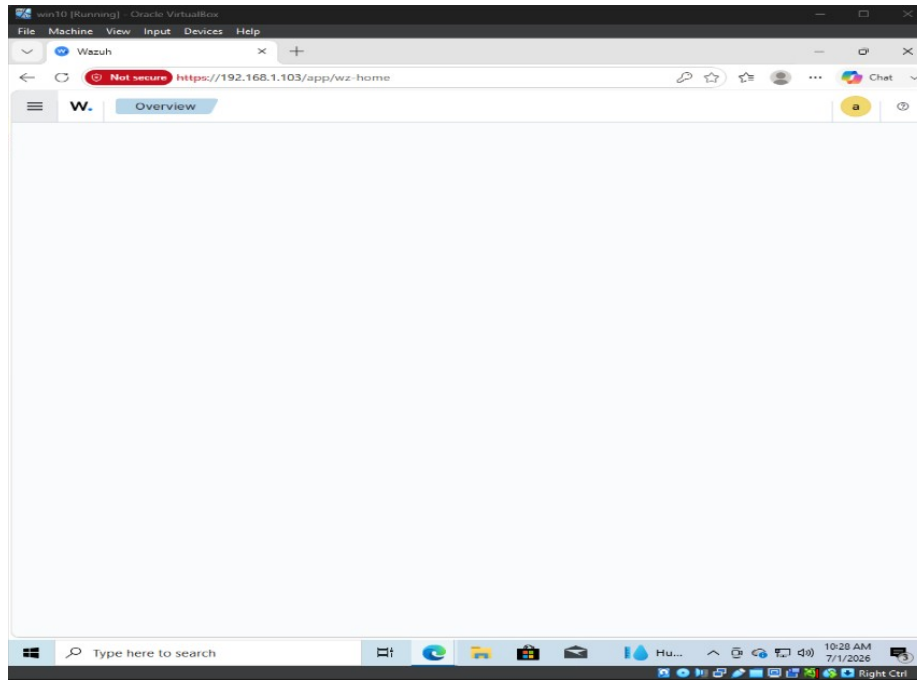
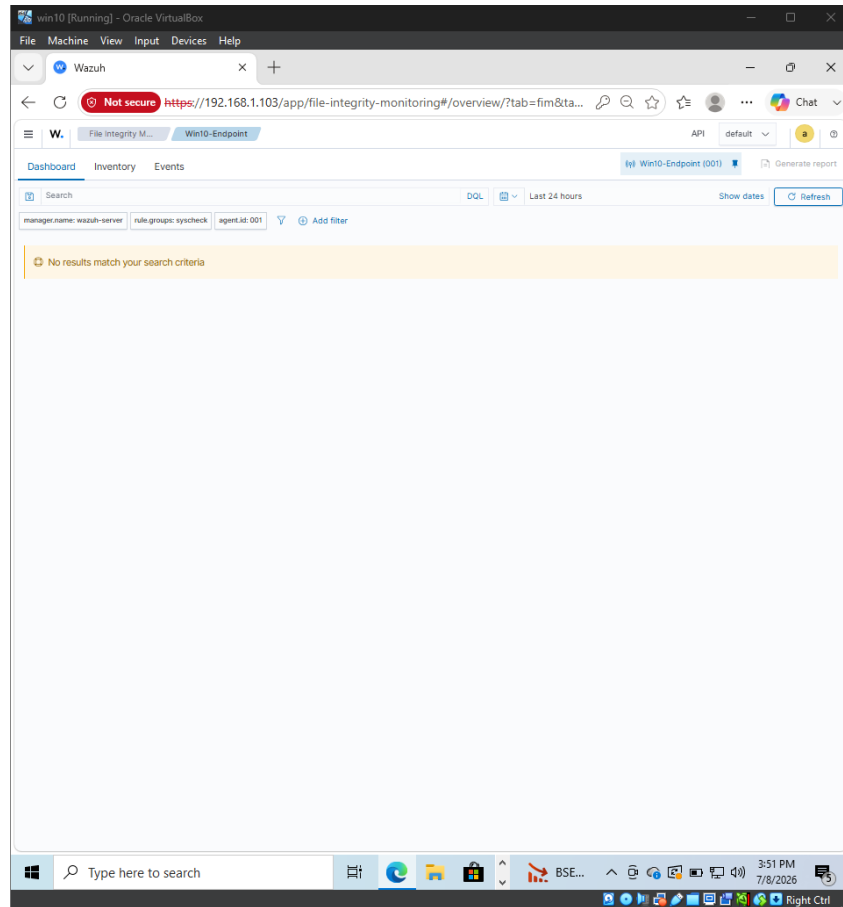


Figure 5.4: Wazuh Agent Connectivity Verification

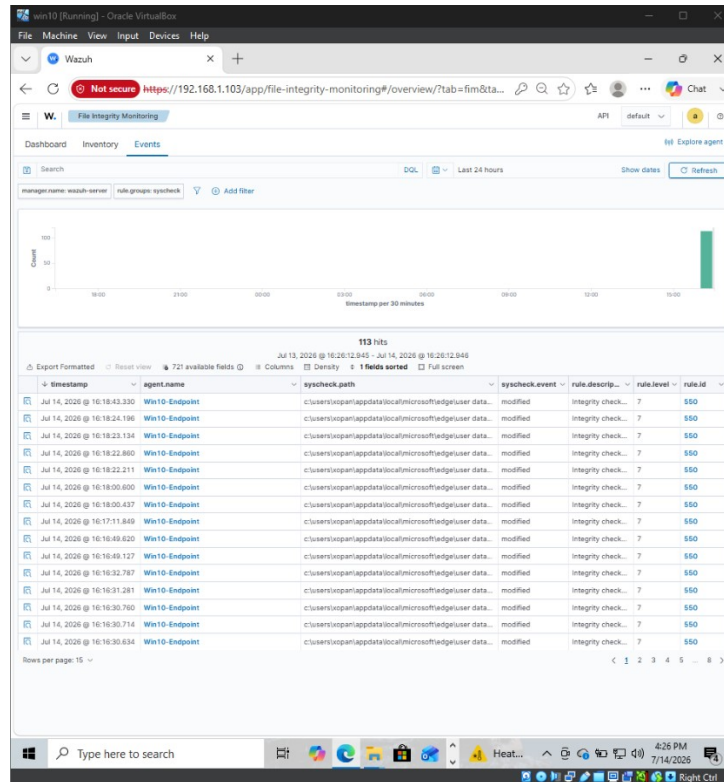
```
Wazuh v4.14.3 OVA [Running] - Oracle VirtualBox
File Machine View Input Devices Help
WAZUH Open Source Security Platform
https://wazuh.com
[wazuh-user@wazuh-server ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:2f:dc:22 brd ff:ff:ff:ff:ff:ff
   altname emp0s17
   inet 192.168.1.103/24 metric 1024 brd 192.168.1.255 scope global dynamic eth0
       valid_lft 6884sec preferred_lft 6884sec
   inet6 fe80::a00:27ff:fe2f:dc22/64 scope link proto kernel_l1
       valid_lft forever preferred_lft forever
[wazuh-user@wazuh-server ~]$ ping -c 4 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data:
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=7.15 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=4.16 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=4.53 ms
64 bytes from 192.168.1.1: icmp_seq=4 ttl=64 time=3.95 ms
--- 192.168.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3011ms
rtt min/avg/max/mdev = 3.948/4.947/7.147/1.287 ms
[wazuh-user@wazuh-server ~]$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=254 time=24.2 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=254 time=19.0 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=254 time=21.6 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=254 time=20.3 ms
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3021ms
rtt min/avg/max/mdev = 19.965/21.257/24.163/1.927 ms
[wazuh-user@wazuh-server ~]$
```

5.5 File Integrity Monitoring (FIM)

File Integrity Monitoring (FIM) was configured in Wazuh to monitor changes to critical files and directories on the Windows 10 endpoint. When a monitored file was modified, Wazuh generated an alert, demonstrating its capability to detect unauthorized file changes.



After modifying a monitored file, Wazuh successfully detected the change and generated a security alert.



5.6 Network Scan Detection

An Nmap operating system detection scan was performed from the Kali Linux virtual machine against the Windows 10 endpoint. Wazuh successfully detected the reconnaissance activity and generated corresponding alerts.

```

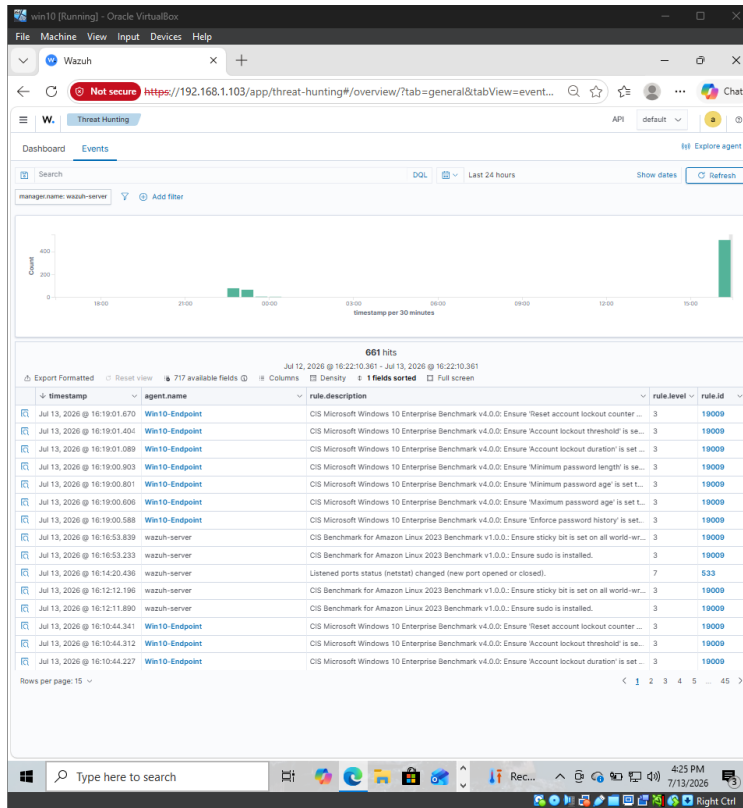
kali@kali:~$ nmap -sV -p- -iL 192.168.1.100
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-15 03:15 EDT
Nmap scan report for 192.168.1.100
Host is up (0.000ms latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds  Microsoft Windows [adsis]
MAC Address: 08:00:27:1E:31:A2 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 11.91 seconds

kali@kali:~$ nmap -sV -p- -iL 192.168.1.100
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-15 04:35 EDT
Nmap scan report for 192.168.1.100
Host is up (0.000ms latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds  Microsoft Windows [adsis]
MAC Address: 08:00:27:1E:31:A2 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running (JUST GUESSING): Microsoft Windows 10[201911] (97%)
OS CPE: cpe:/o:microsoft:windows_10 cpe:/o:microsoft:windows_server_2019 cpe:/o:microsoft:windows_11
Aggressive OS guesses: Microsoft Windows 10 20H2 - 21H1 (97%), Microsoft Windows 10 1809 (94%), Microsoft Windows 10
1909 - 2004 (91%), Windows Server 2019 (91%), Microsoft Windows 11 (91%), Microsoft Windows 10 1909 (90%), Microsof
t Windows 10 1809 (89%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

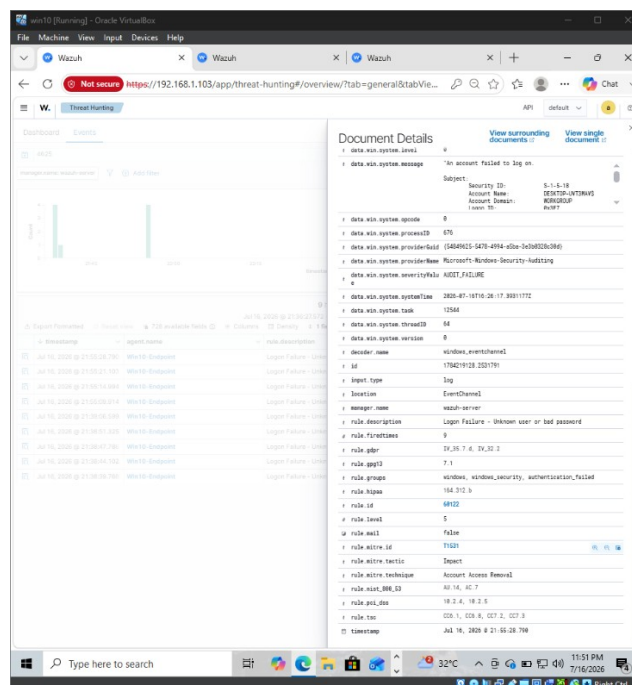
OS detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 25.82 seconds

```



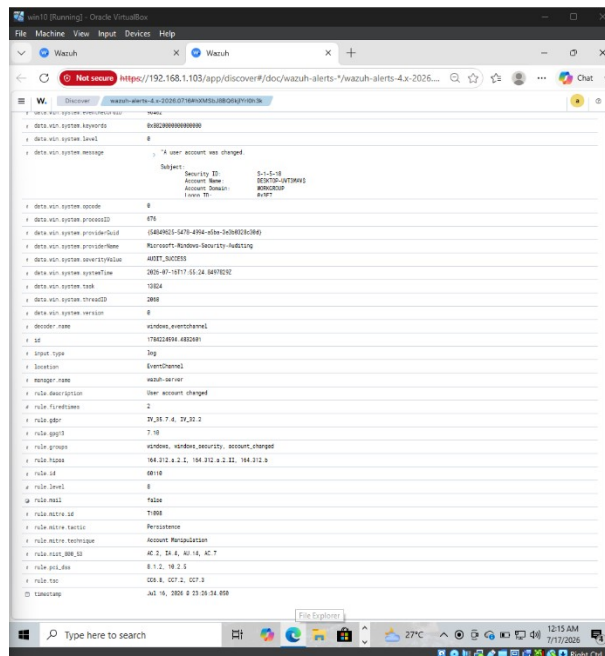
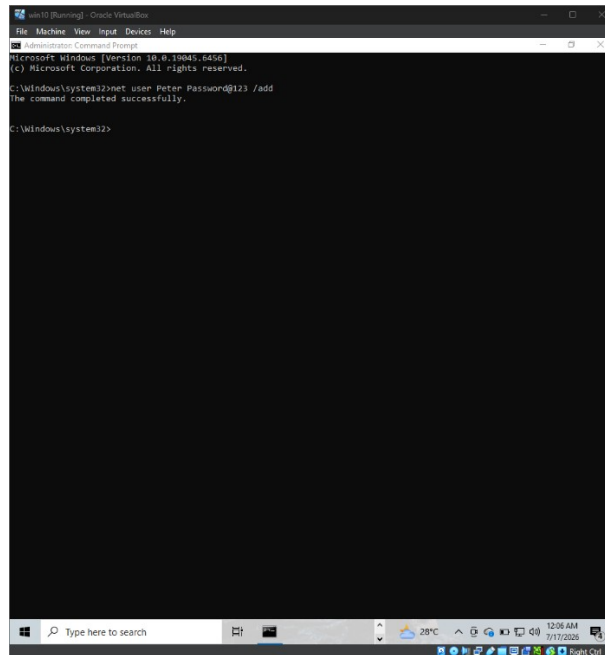
5.7 Failed Login Detection

Multiple failed login attempts were generated on the Windows 10 endpoint. Wazuh detected the authentication failures and mapped the event to the appropriate MITRE ATT&CK technique.

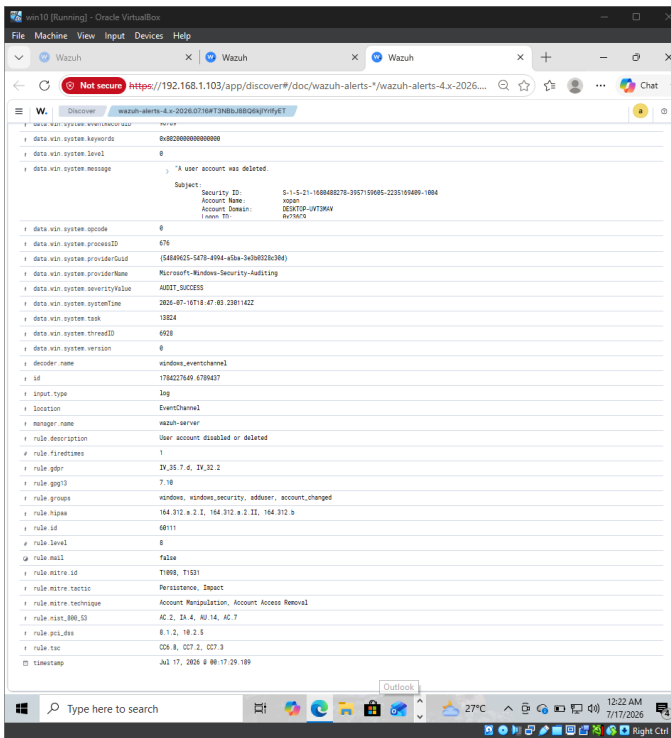


5.8 User Account Monitoring

A local user account was created on the Windows 10 endpoint. Wazuh detected the account creation event and generated an alert.



The previously created user account was deleted. Wazuh successfully detected the account deletion event and generated a corresponding alert.



RESULTS AND DISCUSSION

6.1 Introduction

The implemented Home Network Security Information and Event Management (SIEM) system was evaluated by configuring the virtual environment, deploying the Wazuh platform, enrolling the Windows 10 endpoint, and performing controlled attack simulations using Kali Linux. The results confirmed that the system successfully collected, analyzed, and displayed security events through the Wazuh Dashboard, enabling centralized monitoring and threat detection.

6.2 Experimental Results

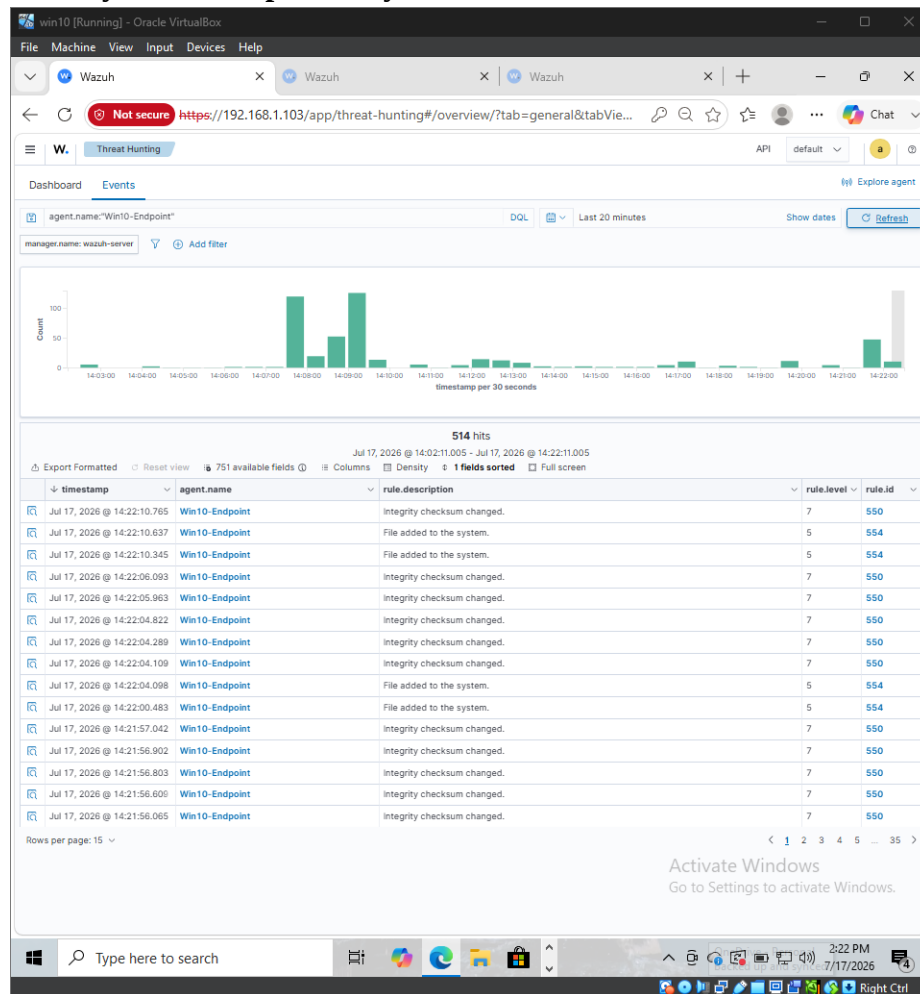
After completing the implementation, all virtual machines communicated successfully through the configured VirtualBox Internal Network. The Wazuh Manager, Wazuh Indexer, and Wazuh Dashboard services operated correctly, while the Windows 10 endpoint forwarded security logs through the Wazuh Agent.

The Wazuh Dashboard displayed endpoint status, File Integrity Monitoring (FIM) events, failed login alerts, network scan detections, user account activity, and other collected security events. The documented attack simulations performed from the Kali Linux virtual machine were detected and analyzed through the Wazuh Dashboard, demonstrating the system's capability for centralized security monitoring and threat analysis.

Table 6.1 Functional Testing Results

Test Case	Expected Result	Observed Result	Status
Wazuh Server Deployment	Wazuh services start successfully	Services started successfully	Pass
Wazuh Agent Registration	Windows endpoint connects to Wazuh Server	Agent registered successfully	Pass
Dashboard Accessibility	Wazuh Dashboard opens correctly	Dashboard accessed successfully	Pass
File Integrity Monitoring (FIM)	File changes generate alerts	Alerts detected successfully	Pass
Failed Login Detection	Failed login attempts generate alerts	Alerts generated successfully	Pass
Network Scan Detection	Nmap scan is detected by Wazuh	Scan detected successfully	Pass
User Account Monitoring	User creation/deletion generates alerts	Alerts generated successfully	Pass
Threat Hunting	Security events available for analysis	Events displayed successfully	Pass

Figure 6.1: Security Events Captured by the Wazuh Dashboard



6.3 Discussion

The results demonstrate that the proposed Home Network Security Information and Event Management (SIEM) system was successfully deployed and that the documented monitoring and detection tests operated as intended. The Wazuh platform effectively collected and analyzed security events generated by the Windows 10 endpoint through the Wazuh Agent, while the virtual network enabled communication between pfSense, Wazuh Server, Windows 10, and Kali Linux.

Attack simulations, including network scanning, failed login attempts, file integrity monitoring, and user account management, were successfully detected and displayed on the Wazuh Dashboard. The generated alerts and the documented MITRE ATT&CK mapping demonstrated the system's ability to identify and analyze selected security events.

Overall, the implementation validated the effectiveness of Wazuh as an open-source SIEM solution for centralized log management, real-time security monitoring, threat detection, and incident analysis within a virtual Security Operations Center (SOC) environment.

Table 6.2: Summary of Security Monitoring Results

Parameter	Observation
Host Platform	Windows 11 with Oracle VirtualBox
Virtual Machines	pfSense, Wazuh Server, Windows 10, Kali Linux
SIEM Platform	Wazuh 4.14.3
Endpoint Monitoring	Wazuh Agent and Sysmon
Dashboard Status	Successfully Accessible
File Integrity Monitoring	Successfully Detected
Failed Login Detection	Successfully Detected
Network Scan Detection	Successfully Detected
User Account Monitoring	Successfully Detected
Threat Hunting	Successfully Performed

Figure 6.2: Detailed Alert Analysis

win10 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Wazuh Wazuh

Not secure https://192.168.1.103/app/discover#/doc/wazuh-alerts-*/wazuh-alerts-4.x-2026...

Discover wazuh-alerts-4.x-2026.07.17#E4xGb58BVJzbMTIQv2ik

Table JSON

@timestamp	Jul 17, 2026 @ 14:22:10.765
_index	wazuh-alerts-4.x-2026.07.17
agent.id	001
agent.ip	192.168.1.100
agent.name	Win10-Endpoint
decoder.name	syscheck_integrity_changed
full_log	File 'c:\users\xopan\appdata\local\packages\microsoft.windows.search_cw5n1h2txyewy\localstate\ebwebview\default\cache\cache_data\index' modified Mode: realtime Changed attributes: md5, sha1, sha256 Old md5sum was: '43efc6cc9229f9f3715865675e2655' New md5sum is: 'dc8a33a78b2427f9f3715865675e2655' Old sha1sum was: '99c10a89d8a6d96df18fec583d462db0e27e392' New sha1sum is: 'bfa0a36704dd8b6c50151a4a08d86d3106b57d302a'
id	1784278330_6126586
input.type	log
location	syscheck
manager.name	wazuh-server
rule.description	Integrity checksum changed.
rule.firedtimes	319
rule.gdpr	II.5.1.f
rule.gpg13	4.11
rule.groups	ossec, syscheck, syscheck_entry_modified, syscheck_file
rule.hipaa	164.312.c.1, 164.312.c.2
rule.id	550
rule.level	7
rule.mail	false
rule.mitre.id	T1565.001
rule.mitre.tactic	Impact
rule.mitre.technique	Stored Data Manipulation
rule.nist_800_53	SI.7
rule.pci_dss	11.5
rule.tsc	PI1.4, PI1.5, CO6.1, CO6.8, CO7.2, CO7.3
syscheck.attrs_after	ARCHIVE
syscheck.changed_attributes	md5, sha1, sha256
syscheck.event	modified
syscheck.md5_after	dc8a33a78b2427f9f3715865675e2655
syscheck.md5_before	43efc6cc9229f9f3715865675e2655
syscheck.mode	realtime
syscheck.mtime_after	Jul 17, 2026 @ 09:52:51.000

Activate Windows
Go to Settings to activate Windows.

Type here to search

30°C

2:24 PM
7/17/2026

Right Ctrl

Figure 6.3: MITRE ATT&CK Mapping of Detected Threats

The screenshot displays the Wazuh web interface, showing a discovered threat rule. The rule is mapped to MITRE ATT&CK T1565.001 (Impact) and T1565.001 (Stored Data Manipulation). The rule description is 'Integrity checksum changed.' and it is associated with the syscheck module.

Property	Value
location	syscheck
manager.name	wazuh-server
rule.description	Integrity checksum changed.
rule.firedtimes	319
rule.gdpr	II.5.1.f
rule.gpg13	4.11
rule.groups	ossec, syscheck, syscheck_entry_modified, syscheck_file
rule.hipaa	164.312.c.1, 164.312.c.2
rule.id	558
rule.level	7
rule.mail	false
rule.mitre.id	T1565.001
rule.mitre.tactic	Impact
rule.mitre.technique	Stored Data Manipulation
rule.nist_800_53	SI.7
rule.pci_dss	11.5
rule.tsc	PI1.4, PI1.5, CC6.1, CC6.8, CC7.2, CC7.3
syscheck.attrs_after	ARCHIVE
syscheck.changed_attributes	md5, sha1, sha256
syscheck.event	modified
syscheck.md5_after	dc8a33a78b2427f9f3715865675e2655
syscheck.md5_before	43ef6cc9229fdfe13b7b9fb418ecb5e
syscheck.mode	realtime
syscheck.mtime_after	Jul 17, 2026 @ 09:52:51.000
syscheck.path	c:\users\xopan\appdata\local\packages\microsoft.windows.search_cw5n1h2txyewy\localstate\ebwebview\default\cache\cache_data\index
syscheck.sha1_after	b6963679dd0bc659151a498486d3196b57d392e
syscheck.sha1_before	99c10a89d8e6d96df18fec583d462db0e27e392
syscheck.sha256_after	eb47bc40d0563c5f41b2a50984830115418b583c1bfc951713ebf0d9bdd78107
syscheck.sha256_before	18221be08288c6a6323e0808b8d0dfb2c983e058a13dc553d7c699fd994c846c1
syscheck.size_after	262,512
syscheck.win_perm_after.allowed	DELETE, READ_CONTROL, SYNCHRONIZE, READ_DATA, WRITE_DATA, APPEND_DATA, READ_EA, WRITE_EA, EXECUTE, READ_ATTRIBUTES, WRITE_ATTRIBUTES, DELETE, READ_CONTROL, WRITE_DAC, WRITE_OWNER, SYNCHRONIZE, READ_DATA, WRITE_DATA, APPEND_DATA, READ_EA, WRITE_EA, EXECUTE, READ_ATTRIBUTES, WRITE_ATTRIBUTES, DELETE, READ_CONTROL, WRITE_DAC, WRITE_OWNER, SYNCHRONIZE, READ_DATA, WRITE_DATA, APPEND_DATA, READ_EA, WRITE_EA, EXECUTE, READ_ATTRIBUTES, WRITE_ATTRIBUTES, DELETE, READ_CONTROL, WRITE_DAC, WRITE_OWNER, SYNCHRONIZE, READ_DATA, WRITE_DATA, APPEND_DATA, READ_EA, WRITE_EA, EXECUTE, READ_ATTRIBUTES, WRITE_ATTRIBUTES
syscheck.win_perm_after.name	S-1-15-3-1024-1528657515-1944437972-2795272136-1227674495-293963776-353393192-4060142787-1908764839, S-1-15-3-536077884-713174666-1066051701-3219990555-339840825-1966734348-1611281757, xopan, SYSTEM, Administrators
timestamp	Jul 17, 2026 @ 14:22:10.765

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The objective of this project was to design and implement a Home Network Security Information and Event Management (SIEM) system using Wazuh and pfSense for centralized security monitoring and threat detection. The system was successfully deployed in a virtualized environment using Oracle VirtualBox, consisting of pfSense Firewall, Wazuh Server, Windows 10 Endpoint, and Kali Linux.

The implemented SIEM solution successfully collected, analyzed, and displayed the documented security events generated by the monitored endpoint. Security monitoring features such as File Integrity Monitoring (FIM), failed login detection, network scan detection, user account monitoring, and threat hunting were successfully demonstrated through the Wazuh Dashboard. The implementation also validated the effectiveness of Wazuh as an open-source SIEM platform for centralized log management, real-time security monitoring, incident analysis, and visualization of security events.

Overall, the project achieved its objectives by providing a practical Security Operations Center (SOC) environment for cybersecurity monitoring, attack detection, and security event analysis using open-source technologies.

7.2 Future Scope

Although the implemented SIEM system successfully achieved its objectives, several enhancements can further improve its capabilities and practical applications.

Future work may include integrating additional security tools such as intrusion detection and prevention systems (IDS/IPS), endpoint detection and response (EDR) solutions, and threat intelligence feeds to improve threat detection accuracy. Automated incident response using Security Orchestration, Automation, and Response (SOAR) platforms can also be implemented to reduce response time during security incidents.

The system can be further enhanced by deploying it in cloud environments such as Microsoft Azure, Amazon Web Services (AWS), or Google Cloud Platform (GCP) to improve scalability and support enterprise-level security monitoring. Additional log sources, including Linux servers, web servers, network devices, and cloud services, can also be integrated to provide comprehensive visibility across heterogeneous environments.

With these enhancements, the proposed SIEM system could evolve into a more scalable and intelligent cybersecurity monitoring solution suitable for broader research, education, and organizational security use.

Table 7.1: Future Enhancement Opportunities

Proposed Enhancement	Expected Benefit
Integration with SOAR Platforms	Automated incident response and faster threat remediation
Threat Intelligence Integration	Improved detection of emerging cyber threats
Cloud-Based SIEM Deployment	Enhanced scalability and remote monitoring
Additional Log Source Integration	Comprehensive monitoring of servers, network devices, and cloud services
Endpoint Detection and Response (EDR) Integration	Advanced endpoint monitoring and threat detection
AI/ML-Based Threat Detection	Improved anomaly detection and reduced false positives

7.3 Final Remarks

The implementation of the Home Network Security Information and Event Management (SIEM) system provided valuable practical experience in deploying and managing a virtual cybersecurity monitoring environment. The project enhanced knowledge of SIEM architecture, network security, log management, endpoint monitoring, threat detection, and incident analysis using open-source technologies such as Wazuh, pfSense, Sysmon, and Oracle VirtualBox.

The successful deployment and testing of the system demonstrated the effectiveness of centralized security monitoring in detecting and analyzing various security events, including file integrity changes, failed login attempts, network scans, and user account activities. The practical experience gained through this project has strengthened the understanding of Security Operations Center (SOC) workflows and modern cybersecurity practices.

Overall, this project serves as a strong foundation for further learning and professional development in cybersecurity, particularly in the areas of security monitoring, threat hunting, digital forensics, and incident response. It also demonstrates the practical application of SIEM technologies in improving organizational security and supporting proactive cyber defense strategies.

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